AI for Smarter Patient Care

Transparent Research Prototypes from Structured Clinical Data

CHALLENGE AT A GLANCE

Challenge	**Build a transparent AI prototype that helps people explore, validate, or model structured hospital data.**
Dataset	MIMIC-IV Clinical Database Demo v2.2: deidentified hospital and ICU records for 100 patients.
Choose one track	Structured Patient Timeline; Cohort & Data Quality Explorer; Research-Only Risk or Trajectory Model.
Intended users	Clinical-data researchers, educators, and healthcare data teams—not clinicians making patient-care decisions.
Hackathon scope	A focused, testable proof of concept sized for a short-format hackathon.
Required outcome	A working prototype, reproducible pipeline, evaluation results, and an explicit safety and limitations statement.

Challenge statement

Hospital data are rich but difficult to use. A single admission may span many linked tables containing laboratory results, medication administrations, diagnoses, procedures, transfers, and time-stamped ICU observations. Before these data can support credible research, teams must reconstruct context, check data quality, prevent leakage, and communicate uncertainty.

Your challenge is to build an AI-powered prototype that makes the supplied **structured** clinical data easier to inspect, validate, or study. The strongest projects will solve one well-defined problem, use AI for a necessary part of the solution, preserve a clear trail back to the source data, and evaluate performance honestly.

This is a research and education challenge. The dataset is too small and narrow to establish clinical effectiveness, safety, generalizability, or improvements in patient outcomes. Projects must not provide diagnosis, treatment, triage, or emergency guidance.

Supported tracks

Choose **one primary track**. Projects may include supporting features from another track, but judging will focus on the selected track and its evaluation criteria.

Track 1 — Structured Patient Timeline & Evidence Retrieval

Build a tool that reconstructs an encounter or ICU stay from relational tables and helps a user find verifiable facts in the structured record.

Possible prototypes include:

1. A time-ordered patient journey showing admissions, transfers, tests, medications, procedures, and ICU observations.
2. A structured-data question-answering interface that answers only when supporting rows are available.
3. An event summarizer that groups high-volume observations without hiding clinically relevant source data or uncertainty.

Every patient-level statement must link to its source table, field, identifier, and time. The supplied dataset contains **no free-text clinical notes**; do not present structured-code labels or generated prose as if they were clinician-authored notes.

Track 2 — Cohort & Data Quality Explorer

Build a tool that helps researchers define a cohort and identify whether the underlying data are fit for a stated analysis.

Possible prototypes include:

1. Natural-language-to-structured cohort queries with visible inclusion and exclusion logic.
2. Detection of missing, duplicated, inconsistent, implausible, or temporally misaligned records.
3. An explainable explorer for measurement coverage, unit variation, coding patterns, and data provenance.

The product must distinguish a true clinical finding from a data-quality flag. It must not silently "correct" or delete source records. Any suggested correction should be reversible and supported by a documented rule.

Track 3 — Research-Only Risk or Trajectory Model

Build a narrowly defined retrospective modeling pipeline using a structured outcome that is present often enough in the demo to evaluate.

Examples include a research prototype for a length-of-stay category, a future measurement threshold, or another clearly defined hospital or ICU trajectory. Teams must state the prediction question, eligible population, index time, prediction horizon, outcome definition, and intended research use before modeling.

The demo is not large enough to validate mortality, readmission, deterioration, treatment-response, or population-health models. A technically correct finding may be that the chosen endpoint is not viable in this sample. Treatment recommendations, causal claims, and deployment claims are out of scope.

Supplied dataset and rules

The challenge uses the **MIMIC-IV Clinical Database Demo v2.2**, an openly available subset of MIMIC-IV.

The supplied data:

1. Cover **100 patients** from one tertiary academic medical center in Boston, USA.
2. Contain retrospective, deidentified hospital and ICU data organized as relational CSV tables.
3. Include structured information such as demographics, admissions, transfers, diagnoses, procedures, laboratory measurements, medication-related events, and ICU observations, subject to table coverage and missingness.
4. Use deidentified identifiers and date shifting. Calendar dates, seasonality, and real-world chronology across different patients must not be inferred.
5. Exclude free-text clinical notes. The separately published MIMIC-IV-Note dataset is not part of this challenge.

These records are a small, non-representative educational sample. They are not sufficient to establish clinical validity, subgroup fairness, treatment effectiveness, hospital-wide operations, or performance in another institution or population.

Challenge rules:

1. Use the organizer-supplied frozen copy of MIMIC-IV Demo v2.2 as the primary patient-level dataset.
2. Public documentation, code systems, and non-patient reference data may be used if their source and version are cited.
3. Pretrained models and synthetic test cases are permitted if clearly identified. Synthetic data must remain separate from real demo records and may not be used to imply real-world clinical performance.
4. Do not add external patient-level data unless the organizers expressly provide it to every team under the same terms.
5. Follow the PhysioNet licence and attribution requirements. Do not attempt reidentification or upload patient-level rows to a service whose terms do not permit that use.
6. Clearly disclose all models, external services, generated data, manual labels, and human-authored rules.

Required deliverables

Each team must submit:

1. **Working prototype** — An end-to-end demonstration of the selected track using the supplied data.
2. **Source code and run instructions** — A reproducible repository or package, including dependencies, configuration, and random seeds where applicable.
3. **Technical summary** — A concise README or brief covering the target user, data flow, AI method, source tables, assumptions, and design choices.
4. **Evaluation report** — Baseline, test protocol, track metrics, results, uncertainty, error examples, and known limitations.
5. **Safety and data statement** — Intended use, prohibited use, data lineage, privacy handling, failure modes, and human-review boundary.
6. **Demo and pitch** — A clear walkthrough of the problem, product, evaluation evidence, and one honest failure case.

Evaluation protocol

All teams must:

1. Evaluate on the exact organizer-supplied dataset version and identify the tables and fields used.
2. Keep all records for a subject_id in the same evaluation fold. Encounters or events from one patient may not appear in both training and test data.
3. For time-dependent tasks, define an index time and use only information available at or before that time. Report and remove label leakage.
4. Compare the AI method with a simple, relevant baseline or rule-based approach.
5. Report sample counts, exclusions, missingness, outcome prevalence where relevant, and uncertainty—not only a single headline score.
6. Include representative errors and show how the product behaves when data are absent, ambiguous, or outside scope.

Because there are only 100 patients, model results are illustrative. Use patient-grouped cross-validation or the organizer-provided split, report fold-level variation or confidence intervals, and avoid claims that small numerical differences establish superiority.

Track-specific metrics

Track	Required metrics	Useful supporting measures
1. Timeline & Retrieval	Structured-fact accuracy; temporal-order accuracy; source-provenance coverage; unsupported-answer or abstention accuracy	Retrieval latency; duplicate-event handling; user task completion
2. Cohort & Data Quality	Cohort-definition correctness; precision, recall, and false-positive rate on documented or organizer-seeded quality issues; reproducibility of results	Issue coverage by table; time to investigate; reversible correction rate
3. Risk or Trajectory	A metric appropriate to the outcome plus a baseline; calibration or absolute error; patient-level uncertainty	PR-AUC and AUROC for classification; MAE/RMSE for regression; sensitivity at a declared threshold

Raw metrics are not comparable across tracks. Judges will assess whether the metric matches the stated problem and whether the evidence supports the team's claims.

Safety and responsible AI

Every prototype must display this notice prominently:

Research and educational prototype only. Not for clinical use. Do not use for diagnosis, treatment, triage, or emergency decisions.

Teams must also:

1. Avoid patient-specific recommendations, treatment rankings, or claims of improved outcomes.
2. Show source provenance, data gaps, and uncertainty for patient-level outputs.
3. Make AI-generated content visually distinguishable from source data.
4. Test hallucination and out-of-scope behavior. A system should abstain when the structured record does not support an answer.
5. Describe where human review is required and prevent automated clinical action.
6. Preserve the original data, log transformations, and make cleaning or imputation reversible.
7. Report relevant subgroup composition while acknowledging that 100 patients cannot support reliable fairness conclusions.
8. Respect the licence, minimize data sent to external services, and never attempt reidentification.

Judging rubric

Category	Weight	What judges will assess
Problem & Impact	**20%**	A precise, data-supported problem; a realistic research or education user; measurable proxy value; and a scope appropriate to the demo dataset.
AI & Data Quality	**25%**	AI is necessary rather than decorative; joins and temporal logic are sound; data lineage is visible; leakage is controlled; and evaluation, baselines, and uncertainty are appropriate.
Working Product	**20%**	A functional end-to-end prototype with usable core workflows, reproducible setup, and sensible handling of missing, malformed, or unsupported inputs.
Safety & Reliability	**15%**	Clear research-only boundaries, provenance, uncertainty, failure testing, human review, privacy and licence compliance, and no unsupported clinical claims.
Innovation	**10%**	A meaningful and well-justified improvement over a simple baseline, with thoughtful use of AI suited to the selected problem.
Pitch & Clarity	**10%**	Clear communication of the problem, data, method, live demonstration, evidence, tradeoffs, limitations, and next validation step.
Total	**100%**	

Resources

1. **MIMIC-IV Clinical Database Demo v2.2 — dataset, documentation, files, licence, and citation:** https://physionet.org/content/mimic-iv-demo/2.2/
2. **Permanent dataset DOI:** https://doi.org/10.13026/dp1f-ex47
3. **Official MIMIC-IV documentation:** https://mimic.mit.edu/docs/IV/
4. **Official schema overview:** https://mimic.mit.edu/docs/IV/about/schema-overview.html
5. **Official MIMIC code repository:** https://github.com/MIT-LCP/mimic-code

Teams should cite the dataset version and any external resources used. The demo page contains the required citation text and current licence information.